Truncating a fiber of a dependent sum: the truncation of the total space is unchanged

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:trunc-in-truncated-sigma' (source lines 629-634, Theorem 7.3.9); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own sentence announcing the fact as a related statement about dependent sums.

Editorial note. The retained counter values (chapter 7, section 3, theorem offset 8) make the retained theorem print as Theorem 7.3.9, matching the original book numbering.

Editorial note. No retained line refers to a label defined outside the retained span, so the package declares no external reference and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: truncation inside a dependent sum (Theorem 7.3.9).

Allowed standard prerequisites (editorial).

- Induction principle of n-truncation. A section of a family of n-types over trunc n A is determined by its values on |a|_n.. It is used repeatedly at source lines 637 and 645-648 to define maps and homotopies on truncated elements.
- Quasi-inverses give equivalences. A map with a two-sided homotopy inverse is an equivalence.. The proof builds phi and psi with homotopies H and K and concludes the equivalence from them at source lines 643-644.

The following related fact about dependent sums is often useful.

Theorem 7.3.9. *Let $P : A \to \mathcal{U}$ be a family of types. Then there is an equivalence*

$$\left\|\sum_{x:A} \|P(x)\|_n\right\|_n \simeq \left\|\sum_{x:A} P(x)\right\|_n.$$

Proof. We use the induction principle of n-truncation several times to construct functions

$$\varphi : \left\|\sum_{x:A} \|P(x)\|_n\right\|_n \to \left\|\sum_{x:A} P(x)\right\|_n$$

$$\psi : \left\|\sum_{x:A} P(x)\right\|_n \to \left\|\sum_{x:A} \|P(x)\|_n\right\|_n$$

and homotopies $H : \varphi \circ \psi \sim \mathsf{id}$ and $K : \psi \circ \varphi \sim \mathsf{id}$ exhibiting them as quasi-inverses. We define φ by setting $\varphi(|(x, |u|_n)|_n) :\equiv |(x, u)|_n$. We define ψ by setting $\psi(|(x, u)|_n) :\equiv |(x, |u|_n)|_n$. Then we define $H(|(x, u)|_n) :\equiv \mathsf{refl}_{|(x,u)|_n}$ and $K(|(x, |u|_n)|_n) :\equiv \mathsf{refl}_{|(x,|u|_n)|_n}$. □